Hi, everyone. I thought to leave you with a quick summary of what would normally be a revision lecture where we just very quickly run through each of the weeks. And I'll try and highlight the key points that I think have been really important. Obviously, don't take this as the only things that you should be looking at. You'll be expected to know all of the content that we've pointed you to. But I just really wanted to pull out the key aspects that I think are conceptually really important for your work if you choose to continue with anything like human AI or human -computer interaction in your next steps. And I'll also end up with a little bit of information about how the exam structure looks like, similarly to what I've been showing you in one of the LGTs. so i'm hoping that for example week one when you look at back at it now everything should be really obvious what we really need you to to look at is are the definitions of interaction design and how it differs to interaction design goals what are you trying to think about as an interaction designer that humans are really important and how you think about humans as well as being able to kind of work with and apply these concepts of affordances and signifiers, which we spend most of the

time thinking about, and
then also be able to start
thinking about how these
apply within GenAI systems,
which, again, we've started
doing. I would also point you
to the not particularly long
blog that talks about AI
as a design material, because
I think that's a really
lovely way from someone who's
very high up in Microsoft,
a lovely way of kind of
getting you to think through
those concepts and see how
people put them together.
So lecture one should be
relatively obvious to you
by now, but these are the
main things to look at.
In terms of lecture two,
obviously we've spent a
lot of time in the lecture
thinking about what is it
cognitively, how does our thinking
work, how does our
perception work. It's really
important, the perception part
especially in thinking part
is really important for design
because that essentially
mediates how your
participants will engage with
whatever you've created but
then i think the really key
key key piece for you is to
very deeply understand what
mental models mean what's
the difference between the
conceptual and represented
and mental models how do
these relate to the goals
of evaluation and execution,
and then be able to link
it back to, you know, AI as
design material, as affordances,
as signifiers, and so
on. So it's the connections
in between the lectures and
not just knowing definitions,

but being able to apply
a definition on a given
system. So if in the exam, you
will get a screenshot of
something and someone says,
well, on this screen, what
would the likely gulfs of
evaluation and execution
look like and what would you
imagine a mental model of
someone who's seeing this
screen for the first time
will be you need to be able to
really apply these concepts
to a problem like that
rather than just rattle off
some definitions and then just
not be able to do anything
else with it all right and
then obviously the distributed
cognition i think as
you all know now it really
ties in with how mental models
and how perception works
so kind of seeing those
connections and being able
to analyze systems within the
the concepts of you know
offloading whether it's
cognitive or computational
annotations embedding embodiment
is what i mean all of
these things will be quite
important for you to to know
and obviously if someone says
you know describe computer
science social actors
theories and then try and
explain what that theory would
predict about a app that
i've just described for you
that again should be something
you're capable of doing.
All right,
so lecture three is then
really looking at the
usability goals and heuristics,
right? And those are
two long lists, and those
might be the only lists

that you might want to kind
of remember, or at least
be able to remember five
out of six, or eight out
of ten kind of concepts
there. But what I really
want you to look at is
understand why, if we're
talking about control how
that relates to mental
models how that relates to
the affordances and signifiers
and all of these things
all of those connections
between lectures again
become really important
so don't just remember
things in a rote way again
try and think about how
would you use this and how
would you apply it look
at the apps on your phone
and try to kind of think
about how the goals map
onto what you're seeing on
the app how the heuristics
map onto an app that you
really like and an app
Timestamp 5 minutes you think is really clunky
if you've never opened
it again. That's probably
the best way of really
getting a sense whether you
understand the material.
Finally, there's been a
section on guidelines, the
emerging guidelines around
human AI. I don't expect
you to remember everything
which was in those papers,
those long lists, but I
want you to be able to,
again, conceptually understand
how does this change
from some of the basics UX
where the goals are clear
and the underlying technology
is very deterministic.
So have a think
about that and again

try and apply that knowledge to an example or two that you see around you. So lecture four then has moved back into very traditional HCI theory, talking about principles of human-centered designs, the different ways in which people represent the similarities across kind of different life cycles or double diamonds or the IDEO five stages or Stanford eSchool five stages and so on so just having a sense of how these design models look like is really helpful for you and being able to talk about these about their differences and similarities it's also going to be helpful the key part here is as you imagine the difference between a need a human need, or a user need, and a requirement, which is kind of something else, and looking at the system, right? So being able to very clearly and succinctly describe a need and a requirement that might fall out of that need, try and apply that onto systems that you see around you. Again, just be able to apply this rather than just describe in abstract, but then also really try and connect that to the methods that you use. And there's been a long list of methods and there was a lot of reading for you to do this week. So kind of go back over all
Timestamp 7 minutes of that and try to connect it again to the material in

lectures one, two, three,
and be able to especially
focus on interviews, for
example, as one of the ways in
the most common ways in which
people would be collecting
user needs and then trying
to infer requirements.
One of the other pointers
I would point you to is um
is the technique of empathy
maps which we've done
in one of the lgts i think
that might be really helpful
for you to look back at
as well so that you're
aware of it and you remember
because i think it's a it's
a nice example of how
you go from a need to
requirements or how you try
and abstract some of the themes
that come out and then
finally there's been sections
around how kind of user
-centered the AI model
development is, and I think
that's also really important
for you to look at and to
understand quite deeply.
In lecture five,
it's prototyping is about
building things, right?
So that has ended up
being a much more kind
of hands-on experience,
and that's what we've
been doing in LGTs. In
terms of the theory and
things that you should
remember for the exam,
really knowing what a prototype
is, what the function of it is,
what role does it
play in design, how
does this relate to
assumption testings,
and why is that even
important? Those are the
kinds of questions that

you need to kind of feel
very confident about. And
again, if someone says,
I'm thinking of building
this kind of an app,
I have these
assumptions, how would a
prototype look like in
this context? you need
to be able to create
something that's
plausible and feasible
and seems reasonable
and kind of makes
sense, right? So
you need to be
able to apply it.
There are some
hierarchies about
types of prototypes
and, you know,
high fidelity, low fidelity,
and some subsections of
those and the pros and cons.
And again, you should be
able to say, well, if I'm
making this prototype for
this app in this context,
I think I'd be using these
methods because I need it
really quick or I need to be
very precise about the UX
and it needs to be pixel
perfect. So all of these things
are, again, about knowing
what your toolbox is and
then being able to apply it.
So lecture six was
your favorite week,
probably reading week.
there's nothing to really
talk about there, but
I just wanted to put
it in so that no one
thinks we've forgotten it.
And let's move on to week
seven. So that's been
Georgia's work that's been
looking at automation agency
and fairness. And I think

there's been some really
interesting examples that
she's pulled out from very
recent HCI literature
around kind of human AI
collaborations. When we try to
abstract away some of the key
concepts, those are around
control and automation.
And you need to be able
to, again, analyze systems
that someone puts in front
of you in these terms.
It's really important for
you to understand what
fairness is and how it's
defined and what the
challenges are and what
do we know about it being
problematic in the AI
systems that we've had, both
in terms of the deep
learning type as well as the
generative models type
that we have now. And then
probably really important,
being able to, again,
articulate the list of
possible harms. That's another,
maybe third now, list that
might be helpful for you
to remember. So knowing
the types of harms, knowing
their applications, knowing
how they emerge, and be
able to analyze a system to
identify how likely is it
that the harms are there.
those would be quite important
capabilities for you to
kind of pick up as you're
preparing for your exam.
And then finally, the
Timestamp 11 minutes concepts of transparency and
explainability and how they
differ, that's really
important for you to know too.
In terms of the
environmental sustainability,
I think that lecture has less

of kind of theory content.
I think that the key
aspects that you really want
to remember from there
are around the sources of
environmental impact that
AI you know has often been
talked about as potentially
being able to stop
global warming and so on
if we get to a particular
level of AI that will allow
us to make breakthroughs
but at the moment it might
actually have quite strong
energy impacts or
sustainability impacts on its
own so knowing that this
happens and kind of being
able to describe it and
talk about it is important
as well as getting a
broader understanding of how
the calculations that talk
about how much AI use,
how much energy AI uses
and which stages, having
a broader understanding
of what that looks
like is probably really
important for you too.
So lecture nine was, as
you remember, a little bit
of a beast where we tried
to go through evaluation,
which could have
easily been a module
just on its own for
all eight weeks.
But the key pieces
for you to really
make sure you understand
are both, again,
the whys, so the purpose
of the evaluation and
understanding why we have
so many methods and the pros
and cons of each, and
then be able to describe
these methods in the different

hierarchies that we've
kind of outlined within
the lecture. So the
differences between precision
or aiming for precision or
aiming for generalizability
or aiming for realism,
as well as kind of understanding
what does a concrete
or abstract or obtrusive
or non-obtrusive methods
look like. That kind of
understanding will be
quite useful for you, as
well as understanding the
pros and cons of involving
or not involving users.
So, again, it's about here's a
toolkit. There's so many tools.
If I place you in front
of a problem, here's an
app, here's a design brief
you're supposed to sort.
How will you make
choices and can you
argue about the
choices you're making?
That's what I really
want you to learn.
So there are some specific
methods that we went into
detail. So the heuristic
evaluations and think aloud.
You've experienced the think
aloud method in the LGT.
Read up on the heuristic
evaluation. and then obviously
there's a lot that we've
spent on experiments so I
will be expecting you to
be able to maybe not go off
and build a two million
pounds worth of experimental
studies somewhere with
clinical populations but I will
expect you to be able to
discuss the differences between
correlation and causation
and why experiments are
designed to address causation

what are the techniques that they use, what does a hypothesis mean, what does independent, dependent, and controlled variables mean, the concepts of reliability and validity and the different subtypes and how they threaten the causation estimation that we want to create within an experiment, as well as the types of design. So kind of very briefly broad understanding of what between and subjects and within subject design means. That's going to be quite important for you too. So I'm conscious that this is the long lecture. I will say again, there's been the relatively long text from Kronbach that I kept on saying, please read. So I'll say it again, like, please read it. There's a lot of theory and detail there that we didn't get into within the lecture. And this is the one lecture where you really need to read that material quite well, because there's a lot of content there that I need you to think about at your own pace before you turn up to the exam. And then finally, red teaming and field studies. I think that's a continuation. In some sense, it's a continuation of the evaluation, right? Because both red teaming and technology probes are testing something about the systems you built. The key piece is to understand

what appropriation is,
the concept of appropriation,
and how red teaming and
field studies or technology
probes as specific examples
both relate to appropriation
in different ways so for
technology probes i'll
expect you to know what the
definition is what is the
role in design when do you use
it do you use it at the very
start do you use it at the
very end do you use it
somewhere in the middle and how
is it different to kind of
prototypes and um how do
you engage with that technology
slightly or with that
technique slightly differently
and then for red teaming
you should remember what
the definition is why is it
needed within the systems how
it relates to lecture what
was it eight seven um the
one that talks about harms
so how does that relate to
harms of ai systems why do
we need it as well as the
types of potential attacks
as well as the defenses. So
I think the constitutional
classifiers, that piece from
Anthropic, is really helpful
for you to read because it
might be really important
and seems like a really
broadly applicable approach.
Okay, so that would have
been the whole content
of the class. There's
been a reasonable amount,
but I think you've also
been able to really engage
with things practically
within the LGTs, which
I was really happy with.
So now let's just move
on to an example of what

could be in the exam.
So I'll be pointing
to the HCI exam,
which is very similar
in style and design.
And I won't be going
through all these things
in too much detail,
but I expect that
Timestamp 17 minutes you're able to pause
the video and then read.
The first thing that I want
to show here is that, as
I kept on saying over the
last what 15 minutes um
most of what you'll be asked
to do is always showing
that you understand a concept
which is not that hard
but then are able to apply
it in in context so it will
be about um in in in the
hci course we've always
had a context so you're a ux
company that's been tasked
by nhs to design a
vaccination passport app for
the use in sporting venues
and the specific focus for
you is to look at how
well can it work for event
organizers and the NHS or for
the travelers or something
like this right so you
will be given a specific
context for any question and
then you're supposed to be
able to show how you're
able to use all the concepts
that we've talked about
and then apply them to that
specific context so how
does that look in practice
so from the same exam
you might be saying well
okay you need to understand
the needs and challenges
so you might be asked
to create an interview
and then what we're really

trying to look at is
hey do you know how an
interview question looks
like are you able to build
a series of questions
that will help you to
understand what a need
could be so do these
seem to be viable do they
follow the right pattern
are you showing that
you've understood
something about the context
and you're asking kind
of questions that would
allow you to understand
the needs so that's
really what we're
looking at and it's going
to be very similar this
is why we've done the
this is why we for example
done interviews during
the lgt so that model
of how do you build an
interview question so
that you actually get
valuable answers back
that's what we're trying to
do and how are you able
to argue and articulate
why you're asking the
questions you're asking
similarly we when we ask
you about um particular
concepts whether it's mental
models or red teaming or what
want you it will always be
within the context of okay
describe something about
a specific thing such as a
map or a huddle booking one
that would break a mental
model people are likely to
have um and then be clear
about what you think is
broken and then suggest an
alternative design right so
this asks you to show us that
not only you can refer to

the definition but then
when given a specific design
problem you're able to show
that you're able to think
like a designer right so
that you're capable of
applying that content and again
there won't be one right
answer but it's just like
within the lgts if you're
doing something that seems
reasonable and shows that
you understand the concepts
um it's going to be okay and
there's some you know rubric
behind the scenes that
relates to are you using the
Timestamp 20 minutes techniques that we've
talked about in the lecture
and then how do we recognize
which one's good and which
one's bad and then finally
again like the same thing
applies for the evaluation
technique so in this case
it's it says we'll go back
to problem statement that's
been defined before and
assume that your prototyping
has been successful outline
what evaluation metrics
would you be looking at
for your system so remember
that's what you were supposed
to do for LGTs as well and
we're just asking you within
that context to give us
plausible indicators of
success right and through the
HCI lectures we were even
able to show people the
full exam they could see all
the questions at this point
but what what they weren't
able to see was the
specific context and it was
really about even if you know
those questions i unless
you're able to apply them on
the context that doesn't

really help you and i think
a similar model is is is
happening within the exam
here we're not going to be
trying to kind of catch you
out on things that you might
have missed somewhere in
the scripts it's really
about the key concepts and
your ability to apply them
from what i've seen within
the lgts i really hope that
everyone will be more than
capable of doing this i think
you've all been very good
at um kind of seemingly
understanding what we're
asking you to do and why
things work um and i hope that
this will help you kind of
de-risk um the way how you
prepare for an exam so you
know that you don't have
to wrote, learn, and remember
a long list, there might
be like three that you want
to kind of keep in mind
but really
be much more focused on
are you able to apply
these concepts, are you
able to use them because
that's what we want you
to be able to do after you
end, right? You don't
have to remember things,
you need to be able to
apply them. So with this,
all the best of luck,
happy new year if
you're watching it after
the holidays, happy
holidays or holiday
break if you're
watching this sooner.
And yeah, fingers crossed.
I really hope that
all of you will do
exceedingly well on the exam.
All right. It was

lovely teaching you
and I hope to see
you around. Bye.